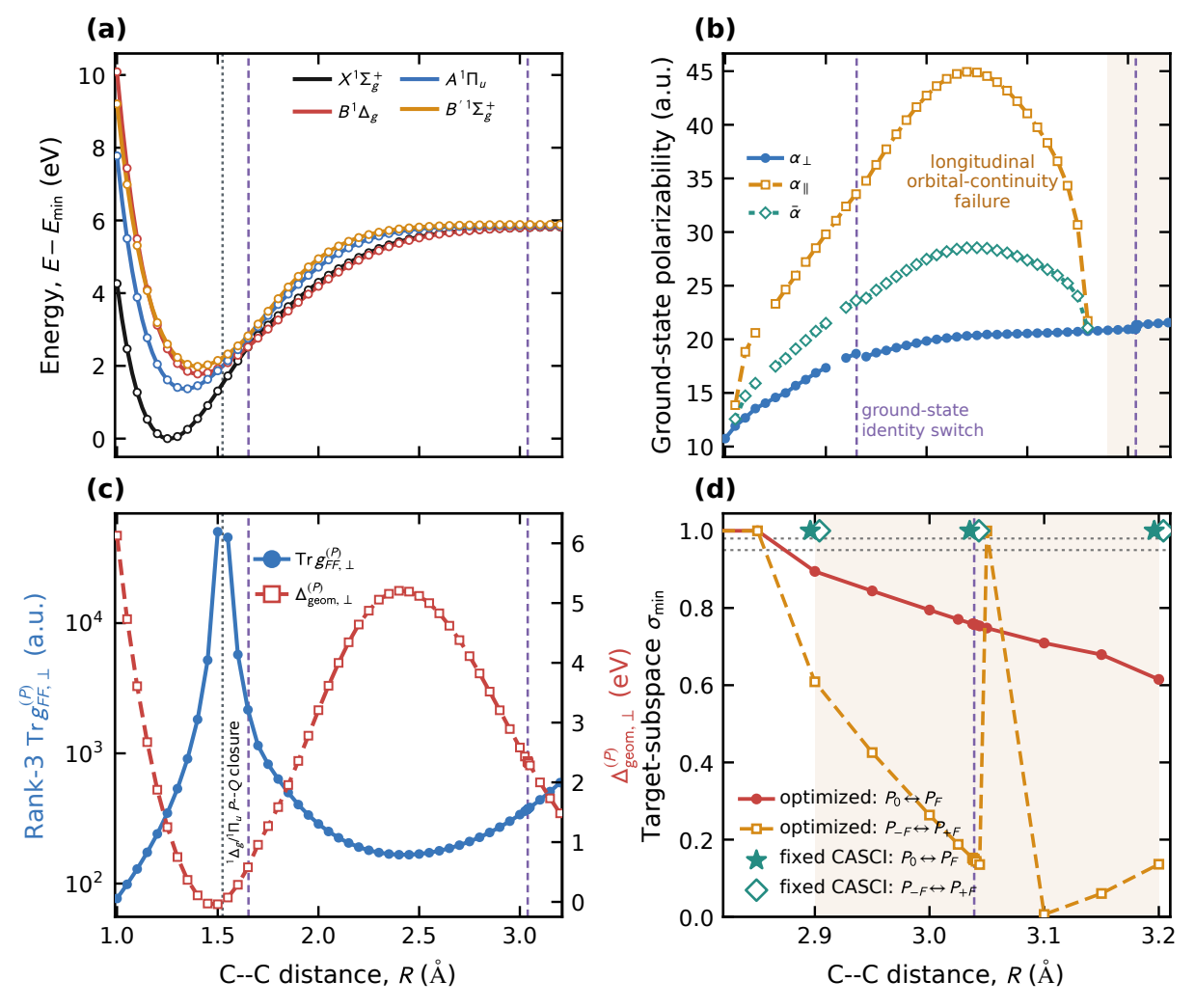

(a)
Energy, $E - E_{\min}$ (eV)
$X^1\Sigma_g^+$
$A^1\Pi_u$
$B^1\Delta_g$
$B'^1\Sigma_g^+$
(b)
Ground-state polarizability (a.u.)
$\alpha_\perp$
$\alpha_\parallel$
$\bar{\alpha}$
longitudinal orbital-continuity failure
ground-state identity switch
(c)
Rank-3 $\mathrm{Tr}\, g^{(P)}_{FF,\perp}$ (a.u.)
$\mathrm{Tr}\, g^{(P)}_{FF,\perp}$
$\Delta^{(P)}_{\mathrm{geom},\perp}$
$\Delta_g/^1\Pi_u$ P--Q closure
C--C distance, R (Å)
(d)
$\Delta^{(P)}_{\mathrm{geom},\perp}$ (eV)
Target-subspace $\sigma_{\min}$
optimized: $P_0 \leftrightarrow P_F$
optimized: $P_{-F} \leftrightarrow P_{+F}$
fixed CASCI: $P_0 \leftrightarrow P_F$
fixed CASCI: $P_{-F} \leftrightarrow P_{+F}$
C--C distance, R (Å)